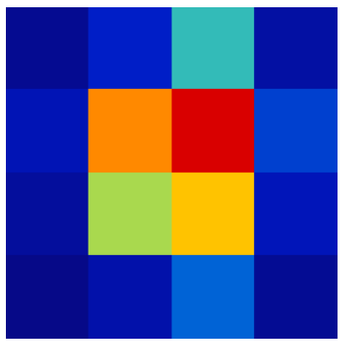
From total Gradient × Input to auditable dual-indexed maps

One registered scalar target s · one observed input x · one frozen computational graph

1 · Total primitive

$x \odot \nabla_x s(x)$



one input-indexed map

i retained · e summed out

2 · Expand by use edge

$\mathcal{L}_{e_Q, i}$

$\mathcal{L}_{e_K, i}$

$\mathcal{L}_{e_V, i}$

joint index $[e, i]$ · conditioned on s

3 · Reconstruct by a valid cut

C_{attn} — Q + K + V

C_{head} — head 1 + ⋯ + head H

C_{res} — identity + branch

$$\sum_{e \in C} \mathcal{L}_{e,i} = x_i \, \partial_i s(x)$$

4 · Ask structural questions

routing ↔ value

identity ↔ branch

space × path × depth

mechanistic comparisons

drop i → edge scalar

drop e → total G×I

drop cut contract → unverified accounting

drop registry s → no target atlas